

Predictive Analysis of Industry Skill Demand using LSPU-SCC Computer Studies Curriculum Data

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OVERVIEW

The project addresses the "skills gap" between academia and industry by aligning the LSPU-SCC curriculum with current job market demands. Currently, the College of Computer Studies (CCS) lacks a data-driven tool to dynamically assess how well its curriculum matches real-time industry needs. This system uses Machine Learning to predict industry demands based on curriculum data and a Knowledge Representation & Reasoning (KRR) layer to provide actionable advisory insights for students and faculty.

General Objective

To bridge the gap between academic programs and industry needs by creating a system that predicts industry skill demands and provides actionable insights to improve graduate employability and curriculum relevance.

Specific Objectives:

1. To develop a multi-class classification model (ML) that analyzes course descriptions to predict relevant job categories (e.g., Data Analyst, Full-Stack Developer).

2. To implement a Rule-Based System (KRR) that acts as an advisory layer to recommend specific skills or curriculum updates based on ML predictions.
3. To develop a web-based application (React + Flask) that visualizes job market alignment and displays recommendations

System Architecture

User Input (Select Program/Specialization) → Machine Learning Model (Skill Extraction & Classification) → KRR Rules (Advisory Logic) → Final Alignment Score → Recommendations (Student/Faculty Advisory)

MACHINE LEARNING COMPONENT (CSST101)

- **Algorithm Used:** Multi-class Classification (Baseline models: Random Forest, Naive Bayes).
- **Dataset Size:** Dynamic (Dependent on scraped job postings and curriculum documents).
- **Model Accuracy:** *To be determined during Week 6 evaluation.*

MACHINE LEARNING PIPELINE

- **Data Collection:** Scraping job postings (JobStreet, LinkedIn) and parsing LSPU-SCC curriculum documents.
- **Data Preprocessing:** Cleaning text and using NLP (NLTK/spaCy) to extract key skill requirements (e.g., Python, SQL).
- **Model Training:** Training baseline models to classify curriculum data against job market categories.
- **Model Evaluation:** Evaluating model performance in predicting job roles (e.g., "Data Analyst," "QA Engineer").
- **Model Deployment:** Integration into a web interface using React for the frontend.

DATASET DESCRIPTION

- **Dataset Type:** Unstructured Text (Curriculum Syllabi) and Web Scraped Data (Job Descriptions).
- **Number of Records:** *Variable* (Aggregated from local and international job portals).
- **Target Variable:** Job Categories (e.g., Data Analyst, Full-Stack Developer, Cybersecurity Specialist)

KNOWLEDGE REPRESENTATION & REASONING (CSST102)

1. **Job Role Prediction Rule**
IF curriculum skills are extracted
AND skills match trained job-role patterns
THEN predicted_job_role = *most relevant job role with highest accuracy*
2. **Skill Gap Identification Rule**
IF curriculum skills are mapped to the Knowledge Graph
AND required industry skills are missing
THEN skill_gap = *identified missing or weak skills*
3. **Curriculum Recommendation Rule**
IF skill gaps are identified
AND target job role requirements are known
THEN curriculum_update = *recommended subjects, tools, or competencies*
4. **Student Upskilling Recommendation Rule**
IF student skill profile is compared with target job role
AND gaps exist in the Knowledge Graph
THEN upskilling_path = *recommended courses, certifications, or training*

HYBRID DECISION LOGIC

The system uses hybrid decision logic by combining Machine Learning predictions and symbolic rule-based reasoning. Machine Learning models handle uncertainty by predicting suitable job roles, while the Knowledge Graph and IF–THEN rules provide structured reasoning to identify skill gaps and generate explainable recommendations.

SYSTEM FEATURES

☑ Curriculum–Job Role Prediction

Uses a **Machine Learning Job Role Classifier** to predict the most relevant IT industry roles based on skills extracted from the LSPU-SCC Computer Studies curriculum.

☑ Semantic Skill Gap Identification

Employs an **Industry Knowledge Graph** to identify missing or weak skills through semantic relationships rather than keyword matching.

☑ Rule-Based Recommendation Engine

Applies **IF–THEN reasoning rules** to recommend:

- Curriculum updates for faculty
- Personalized upskilling paths for students

☑ Interactive Web Interface / API

Provides a **React-based dashboard** and **FastAPI backend** that allow users to:

- Upload curriculum PDF files
- Visualize the industry skill graph
- View job-role match percentages and recommendations

☑ Cloud-Based Development and Evaluation Environment

Uses **Google Colab** and **Jupyter Notebooks** for training, testing, and evaluating NLP pipelines, ML models, and Knowledge Graph reasoning components.

TESTING AND EVALUATION

Test Case	Input Summary	Expected Output
TC-01	Upload LSPU-SCC curriculum PDF	Extracted skills and topics using NLP (spaCy)
TC-02	Extracted skills passed to Job Role Classifier	Predicted IT job roles with confidence scores
TC-03	Skills mapped to Industry Knowledge Graph	Identified semantic skill gaps
TC-04	Detected skill gaps processed by rule engine	Recommended curriculum updates
TC-05	Student skill profile compared with target job role	Suggested upskilling path
TC-06	User accesses interactive dashboard	Visualized skill graph and match percentages

CONCLUSION

This project successfully demonstrates the integration of Machine Learning and Knowledge Representation to solve the "skills gap" problem. By dynamically analyzing industry data, the system provides LSPU-SCC with a tool to ensure its curriculum remains relevant, directly supporting SDG 4 (Quality Education) and SDG 8 (Decent Work).

Group Contribution

1. Ian Patrick Messias – Mathematical, Logical Computation and Documentation
2. Francis Mistika – Model Training and Web application.
3. Dheyne Michael Orlanda – Model Datasets, Documentation, and Papers.

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